NOVACRM · CUSTOMER SUCCESS

Customer Onboarding Automation

Requirements & Workflow Analysis

Prepared for	Rocketlane — Forward Deployed Engineer assessment (Part 1)
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1. Executive summary

NovaCRM's Customer Success team onboards every new customer manually. An Account Executive (AE) emails the CS inbox when a deal closes; a CSM then creates a project from a template, opens a Slack channel, and tracks fifteen tasks across four phases. The process fails in three recurring ways:

1. **Template errors.** The email does not state the plan tier, so the CSM infers it and sometimes applies the wrong onboarding template.
2. **Unverified data migration.** Migration tasks can be marked complete without the customer confirming their data.
3. **Missed escalations.** Escalating stalled work depends on someone noticing it.

This document specifies a multi-agent system that automates the handoff from closed deal to active onboarding project:

- An **Intake & Routing Agent** reads and validates the deal email, and asks the AE for anything missing.
- A **voice agent** calls the AE to confirm the plan tier.
- The system **creates the Rocketlane project** from the correct template (Enterprise, 30 days; Growth, 14 days) and assigns the CSM.
- A **Communication Agent** opens a personalised Slack channel for the customer.
- **Rocketlane automations** escalate overdue tasks and request the customer's sign-off on migrated data.

Guiding principle: deterministic code controls the workflow. Language models are confined to reading and writing text, and never decide whether anything is created. When information is missing or uncertain, the system asks a person rather than guessing.

2. Current-state workflow

2.1 Process

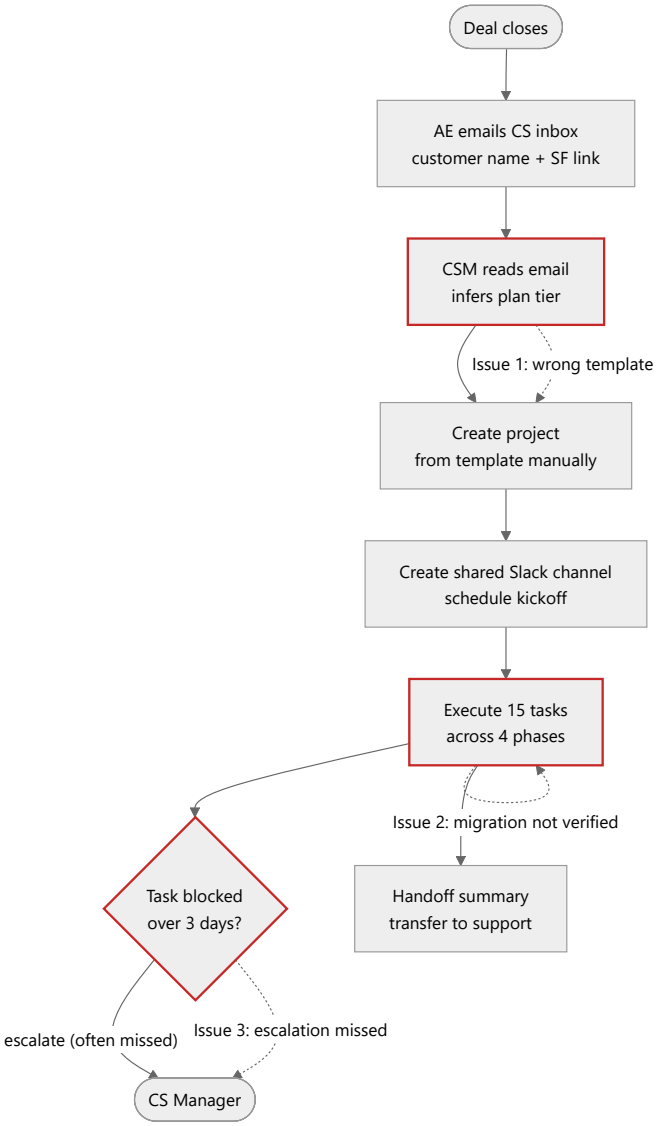


Figure 1. Current-state onboarding workflow and failure points

Step	Owner	Tool	Observed issue
Deal handoff	AE	Email	Unstructured text; plan tier absent; fields occasionally missing
Project creation	CSM	Asana	Template selected manually; Enterprise and Growth confused
Customer channel	CSM	Slack	Repetitive and inconsistent set-up
Phase checklists	CSM, customer	Asana	Migration can be closed without customer verification
Escalation	CSM	None	Relies on memory
Transfer to support	CSM	Document	Manual write-up

2.2 Root-cause analysis

Issue	Root cause	Remedy in the proposed system
Wrong template applied	The tier is not in the email, so it is inferred	Tier **confirmed by phone with a read-back**, checked independently against the call transcript, and **mapped to a template by fixed configuration**. No default tier exists.
Migration closed unverified	Completion is a single click with no customer step	"Customer verifies migrated data" is assigned to the customer and triggers a **formal approval request** in Rocketlane when completed.
Escalations missed	No system enforces them	**Rocketlane automation** notifies the Project Manager at 1 day overdue and the Project Owner at 4 days.
Slow, error-prone set-up	Manual transcription from email	Fields extracted by the Intake Agent **with supporting quotations**, then validated by code.

3. Objectives and scope

Objectives

1. Every closed deal becomes a correctly templated Rocketlane project without manual entry.
2. No project is created on an unconfirmed or ambiguous plan tier.
3. Every customer receives a consistent, personalised Slack channel on day one.
4. Overdue work is escalated automatically, and migrated data is signed off by the customer.
5. Every automated action is traceable: what was done, when, with what input, and why.

In scope: Gmail-triggered intake; voice confirmation of the plan tier; Rocketlane project creation with CSM assignment; the personalised Slack channel; overdue-task escalation and migration sign-off in Rocketlane; guardrails; audit trail; automated tests.

Out of scope for this phase (see §11 roadmap): kickoff-call scheduling, the end-of-onboarding handoff summary, escalation of *blocked* tasks, Salesforce integration, and migration of existing Asana projects.

4. Proposed workflow

4.1 End-to-end sequence

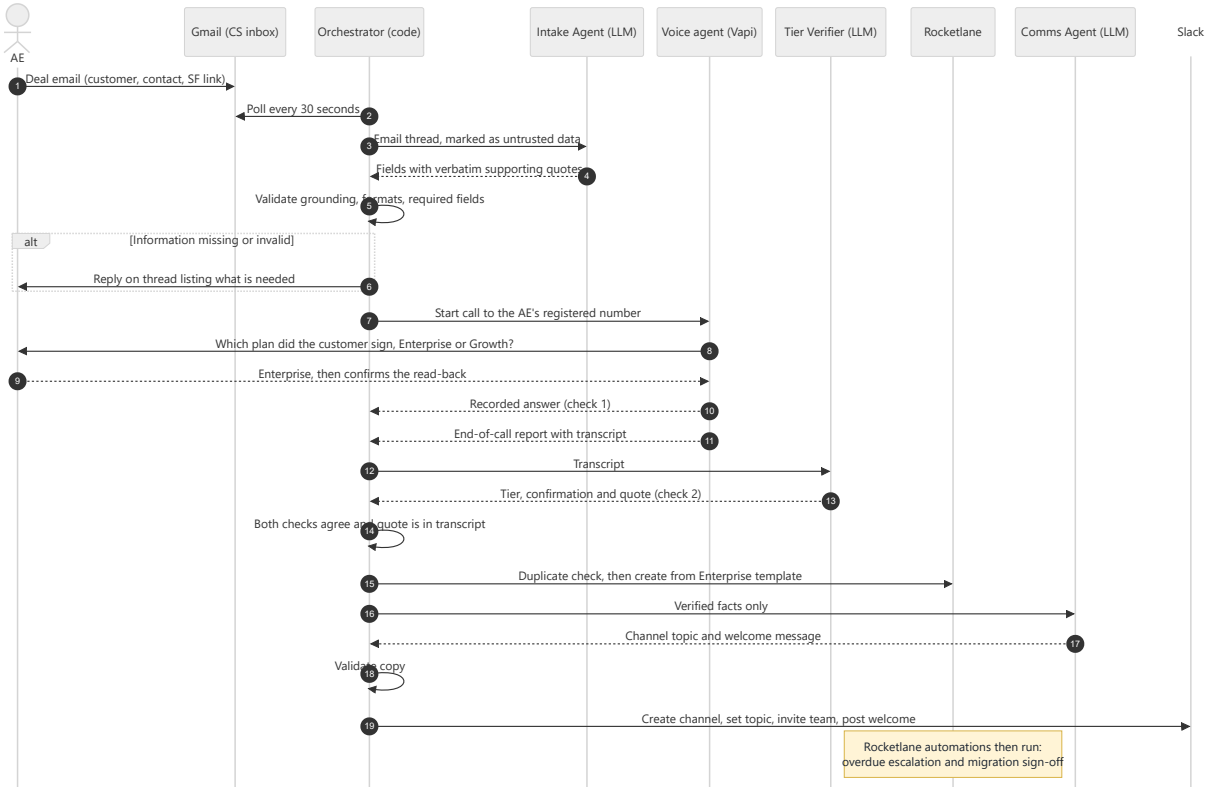


Figure 2. Proposed end-to-end sequence

4.2 Roles and responsibilities

Component	Type	Input	Output	Constraint
Intake & Routing Agent	LLM, OpenAI Agents SDK, no tools	Email thread, wrapped as untrusted data	Customer, contact name and email, AE name, Salesforce link, each with a supporting quote; whether the email is a deal handoff	Cannot infer the tier, invent values, or trigger actions
Voice agent	Vapi assistant on a Twilio number	AE name, customer name	The AE's answer and explicit confirmation, via a single tool call	Records only; makes no decisions
Tier Verifier	LLM, no tools	Call transcript	Tier, confirmation flag, supporting quote	Must agree with the voice agent's record
Orchestrator	Deterministic state machine	Validated agent outputs	All external actions: calls, projects, channels, escalations	Acts only on validated data
Communication Agent	LLM, no tools	Verified facts only; never email text	Channel topic and welcome message	Output validated by code before posting
Overdue escalation	Rocketlane automation	Task due dates	Notifications to Project Manager and Project Owner	Configured in Rocketlane; no custom code

5. Solution architecture

5.1 Component view

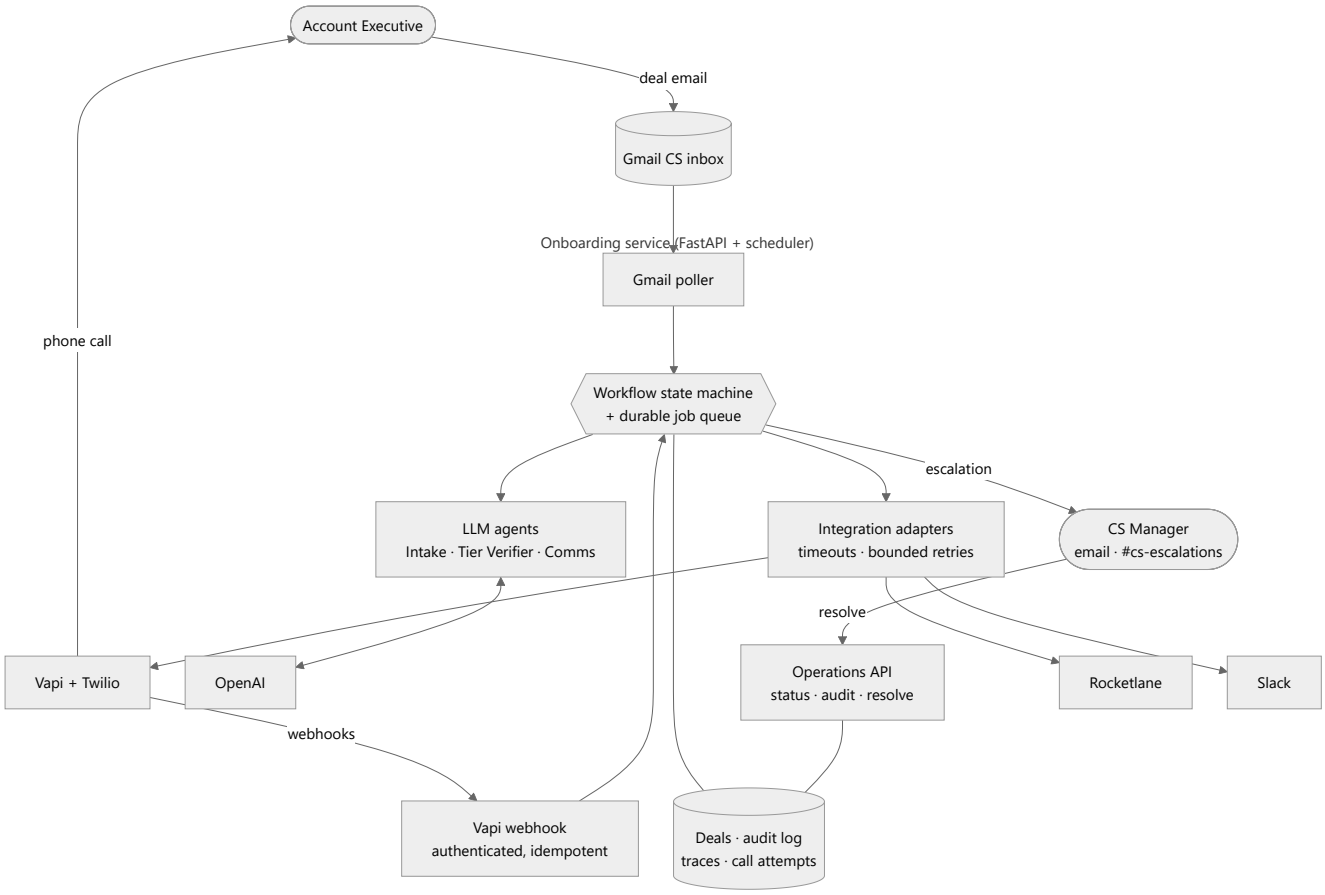


Figure 3. Solution architecture

5.2 Design principles

1. **Deterministic control, LLMs at the edges.** A state machine governs every transition. Language models perform three bounded tasks: interpreting the email, verifying the transcript, and drafting Slack copy.
2. **Agents propose; code executes.** Each agent returns a typed, validated result and has no tools that change anything. Only code calls Rocketlane or Slack.
3. **Escalate rather than guess.** Missing data triggers a request to the AE; an unclear tier triggers a retry, then a human. There is no default tier.
4. **Resilient and idempotent.** Work runs as durable jobs that survive restarts. Each creation step is idempotent: a retry finds the existing record instead of duplicating it.
5. **Replaceable integrations.** Each external service sits behind a single interface, so tests use fakes and production changes are configuration.

5.3 Deal lifecycle

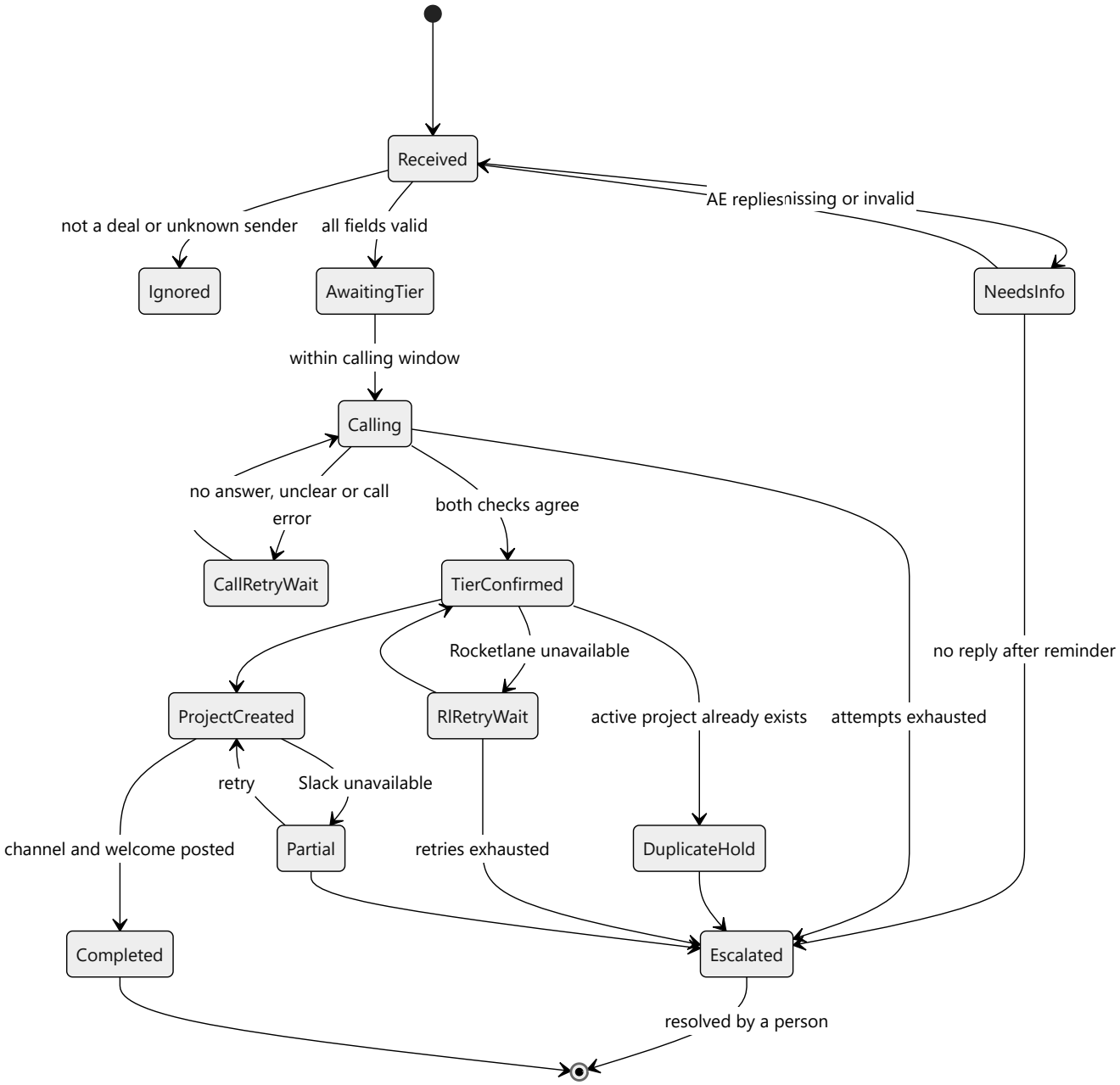


Figure 4. Deal lifecycle

Technology: Python 3.12, FastAPI, OpenAI Agents SDK, Pydantic, SQLite (PostgreSQL in production), APScheduler with a persistent job store, Vapi with Twilio telephony, Rocketlane REST API, Slack Web API.

6. Requirements

6.1 Functional requirements

ID	Requirement	Implementation
FR-1	Detect new deal emails in the CS inbox	Poll every 30 seconds; each message processed exactly once; only allow-listed senders accepted
FR-2	Extract customer name, contact name and email, AE name and Salesforce link	Intake Agent returns each field with a verbatim quote; code verifies both appear in the AE's text
FR-3	Do not proceed on incomplete or invalid data	Reply on the email thread naming the missing items; one reminder; then escalation
FR-4	Confirm the plan tier with the AE by phone	Voice agent calls the AE's registered number, asks for the tier, reads back customer and tier, and requires explicit confirmation
FR-5	Act only on an unambiguous tier	Two independent checks (voice record and transcript verification) must agree, and the supporting quote must appear in the transcript
FR-6	Create the project with the correct template and timeline	**Enterprise:** 30-day template, dedicated CSM assigned round-robin. **Growth:** 14-day template, pooled CSM. Mapping held in configuration.
FR-7	Prevent duplicate projects	A project carrying this deal's reference is adopted; an active project for the same customer holds the deal for review
FR-8	Create a personalised Slack channel	Channel `#onb-<customer>-<id>`; tier-specific topic and welcome naming the customer, tier and CSM; customer invited via Slack Connect
FR-9	Escalate overdue tasks	Rocketlane automation: 1 day overdue → Project Manager; 4 days → Project Owner
FR-10	Obtain customer sign-off on migrated data	Completing "Customer verifies migrated data" sends an approval request to the customer
FR-11	Support human intervention	Escalations by email and in `#cs-escalations`; an operations endpoint to confirm a tier, retry a step, re-read the email or dismiss, with the person and reason recorded

6.2 Non-functional requirements

Area	Requirement
Auditability	Every action records timestamp, actor, input, output and rationale. Agent runs are traced and grouped by deal. Both are available per deal through the operations API.
Reliability	Every external call has a timeout, bounded retries with back-off, and a defined failure outcome. Work survives restarts; a periodic sweep recovers interrupted work.
Idempotency	No duplicate deals, calls, projects, channels or messages, including when a request times out after succeeding.
Security	Webhooks authenticated before the payload is read; operations API protected by a token; secrets held only in environment configuration; email and transcript text treated as untrusted input.

Area	Requirement
Calling etiquette	Calls only within the AE's business hours (configurable window and time zone), maximum three attempts.
Configurability	Business rules (required fields, templates, CSM pools, retry intervals, calling window) are configuration, not code.
Scale	Sized for ~200 customers and a few deals per week. §11 sets out the changes for higher volume.

7. Guardrails and exception handling

Scenario	System behaviour
Email lacks customer name or contact email	Processing pauses; the AE is asked for the specific items; reminder after 24 hours; escalation after 48 hours
Extracted value not present in the email	Rejected: every value and its quote must appear in the AE's text
Non-deal email or unknown sender	Recorded and ignored; no call placed
AE does not answer	Retried after 10 and 30 minutes within the calling window; escalated after three attempts
Ambiguous answer	The voice agent asks once more; if still unclear, recorded as unclear (never mapped to a tier), retried, then escalated
Voice record and transcript disagree	Treated as unclear: retried, then escalated
Rocketlane unavailable	Retried after 1, 5 and 15 minutes, then escalated; success is never assumed
Request timed out but succeeded	The next attempt finds the project by its deal reference and adopts it; no duplicate
Customer already has an active onboarding project	Deal held and escalated; a second project is never created
Slack or the copy-writing model unavailable	Channel set-up resumes from the last completed step; a validated standard message replaces the generated copy
Any escalation	A person is notified with what happened, what was attempted and how to resolve it

8. Rocketlane configuration

8.1 Templates

Both templates share four phases and fifteen tasks and differ in duration.

Phase	Tasks
Kickoff	Review sales handoff · Kickoff call · Agree success criteria and plan
Data Migration	Provide data export · Field mapping sign-off · Sandbox import · **Customer verifies migrated data** · Production import

Phase	Tasks
Configuration	Workspace and user setup · Pipelines and custom fields · Integrations · Admin training
Go-Live	UAT sign-off · Go-live · Handoff to support

	Enterprise	Growth
Template	Enterprise Onboarding (30d)	Growth Onboarding (14d)
CSM model	Dedicated CSM, assigned round-robin	Pooled onboarding team
Project Owner	CS Manager	CS Manager

8.2 Automations

Automation	Trigger	Action
Overdue task escalation	Task becomes overdue (onboarding templates only)	1 day: notify the **Project Manager** (CSM). 4 days: notify the **Project Owner**. Each fires once per task.
Migration sign-off	"Customer verifies migrated data" marked Completed	Request approval from the **Customer**

9. Decisions and assumptions

9.1 Confirmed decisions

Clarifying questions were sent to Karthik Ramaiah (karthikr@rocketlane.com) on 28 October 2026. On 29 October he confirmed the proposed approach for each. All six are implemented as configuration.

#	Question	Decision
Q1	What does a deal email contain: the customer contact email, the AE's phone?	The contact email is in the email body. The AE's phone comes from an AE directory, with the email as fallback. Required fields are configurable.
Q2	How is the dedicated CSM chosen; who are the Project Manager and Owner?	Enterprise: dedicated CSM by round-robin from a pool. Growth: shared onboarding team. CSM is the Project Manager; the CS Manager is the Project Owner.
Q3	Same tasks for both tiers, or different task lists?	Same fifteen tasks; different durations (30 and 14 days).
Q4	Confirm the customer on the call? Retry limit and escalation route?	The call names the customer and the AE confirms the customer and tier together in the read-back. Three attempts, then escalation to the CS Manager.
Q5	Escalate blocked tasks as well as overdue ones?	Overdue only, as specified. Blocked-task escalation is a roadmap item.
Q6	Require customer sign-off on data migration?	Yes, through a Rocketlane approval on the verification task.

9.2 Assumptions

1. Rocketlane replaces Asana for new onboardings; existing Asana projects are not migrated.
2. Only the Enterprise and Growth tiers exist; any other answer is treated as unclear.
3. The tier comes from the AE call, as specified; Salesforce is not queried.
4. AEs are called only during business hours (default Monday to Friday, 09:00 to 19:00 IST).
5. Slack Connect, which invites the customer into the channel, requires a paid Slack plan. In this implementation the invitation is logged rather than sent; the channel, topic and welcome message are live.

10. Verification

An automated suite of 449 offline tests (all external services replaced by fakes) runs with linting and strict type checking before every change.

Assessment requirement	Coverage
Happy path	End-to-end test of every job through its real entry point: email → intake → call → webhooks → tier decision → Enterprise project → personalised channel → completed
Validation	Missing customer name or contact email, values absent from the email, invalid formats, non-deal emails, unknown senders
Template accuracy	Enterprise → 30-day template with dedicated CSM; Growth → 14-day template with pooled CSM
Edge cases	Rocketlane unavailable; existing project; timed-out creation adopted without duplication; unanswered call; ambiguous answer; disagreement between checks; Slack unavailable; repeated webhooks

Each component was also verified against the live services: Gmail intake, a real phone call confirming the tier, project creation in Rocketlane from the correct template, and the personalised Slack channel.

11. Production readiness and roadmap

11.1 Changes for production

Area	Current implementation	Production approach
Slack Connect	Customer invitation logged (paid feature)	Enable on a paid workspace; single configuration flag
Telephony	Twilio trial (trial notice precedes each call)	Paid Twilio account with a local number for the AE's country
Email trigger	Polling every 30 seconds	Gmail push notifications via Pub/Sub
Data store	SQLite, single instance	PostgreSQL, multiple instances; no code change
Secrets	Environment file	Secrets manager with key rotation
Monitoring	Health endpoint, logs, traces	Alerting on escalations and failed jobs; operations dashboard

11.2 Roadmap

1. **Blocked-task escalation:** a Rocketlane automation on tasks blocked for more than three days.

2. **Kickoff scheduling:** propose meeting times from the CSM's calendar in the welcome message.
3. **Handoff summary:** at Go-Live, generate a summary from project and channel history for the support team.
4. **Salesforce integration:** pre-fill the tier from the opportunity, retaining the call as confirmation.
5. **Continuous improvement:** analyse escalation reasons to improve the AE handoff format.